Supplementary Material

1 SUPPLEMENTARY TABLES AND FIGURES

1.1 Tables

Table S1. Comparison of pre- and post-optimization torque prediction fits for BPAs in flexor and extensor configurations on the pinned and biomimetic knees. GoF metrics are RMSE ($N \cdot m$), FVU, and maximum residual ($N \cdot m$).

Configuration		ϕ	l_{rest}, cm	Pre-optimization			Optimized		
				RMSE	FVU	Max. Residual	RMSE	FVU	Max. Residual
Flexor	Pinned	10	48.5	4.871	0.723	9.865	1.608	0.095	3.242
		10	45.7	3.024	0.163	6.753	1.915	0.072	4.695
	Biomimetic	10	41.5	7.448	5.557	13.308	0.680	0.052	1.055
		20	41.5	14.803	3.366	23.715	2.868	0.126	3.550
Extensor	Pinned	10	41.5	4.520	3.972	11.741	2.042	0.844	3.588
		10	45.7	1.999	0.892	4.609	0.896	0.179	1.846
		10	48	0.845	0.232	1.951	0.682	0.151	2.068
	Biomimetic	10	51.8	3.032	4.486	5.587	1.461	1.041	2.755

Table S2. Muscle origin and insertion locations for different models. Origin is in femur reference frame, insertion is in the tibia reference frame.

Muscle	Origin/Insertion	Model	X (m)	Y (m)	Z (m)
Flexor	origin	Pinned knee	-0.075	0.100	0.0328
Flexor	insertion	Pinned knee	-0.05011	-0.045	0.0326
Extensor	origin	Pinned knee	0.030	-0.050	0
Extensor	insertion	Pinned knee	0.0425	-0.076	0
Bifemsh	origin	Gait2392	0.005	-0.211	0.023
Bifemsh	insertion	Gait2392	-0.03	-0.036	0.029
Vastus int.	origin	Gait2392	0.029	-0.192	0.031
Vastus int.	insertion	Gait2392	moving	moving	0.0018
Flexor, 10 mm	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, 10 mm	insertion	Biomimetic knee	-0.0279	-0.046	0.0328
Extensor, 10 mm	origin	Biomimetic knee	0.040	0.035	0
Extensor, 10 mm	insertion	Biomimetic knee	0.021	-0.072	0
Flexor, 20 mm	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, 20 mm	insertion	Biomimetic knee	-0.0279	-0.046	0.0328

1.2 BPA routing and via points

The tables in this section list the routing via points of the BPA configurations reported in the main text. All coordinates are coded in meters in the project's MATLAB sources and are reported here in millimeters, rounded to 0.01 mm. Knee angle θ_k follows the model convention with flexion negative. Femur-side points (path rows before the crossing index) are specified in the femur-referenced instantaneous-center-of-rotation (ICR) frame and are used untransformed at every knee angle, while tibia-side points are specified in the tibia θ_1 frame and are mapped into the ICR frame at each pose with the measured ICR migration, fitted as cubic splines of the tibia-to-ICR offsets over the knee range. The pinned-knee bracket reference points use the knee-ICR and hip frames noted per row, and the pinned-knee route endpoints (origins and insertions) are listed in Table S2.

Table S3. Bracket reference points of the pinned-knee BPA routes, in mm. The pinned flexor route runs from origin to insertion with a compliant bracket at each end, and the pinned extensor route additionally wraps on two fixed arcs ($R_1 = 12\,\mathrm{mm}$, active for $-23^\circ \le \theta_k \le 27^\circ$, and $R_2 = 40\,\mathrm{mm}$, active for $\theta_k \le -23^\circ$).

Configuration	Point	Frame	x (mm)	y (mm)	z (mm)
Pinned flexor	$\mathbf{p}_{bri}$, insertion bracket	knee ICR	-48.11	-107.81	13.80
Pinned flexor	$\mathbf{p}_{br2}$, origin-side second bracket	hip	-52.61	0.00	75.06
Pinned extensor	$\mathbf{p}_{br}$, rib midpoint, medial side	hip	-3.84	-46.44	62.50

Table S4. Routing via points of the biomimetic-knee $10\,\mathrm{mm}$ BPA routes, in mm. The flexor route is a two-point path (crossing index 2), and the extensor route is the eight-point path of the extensor test (crossing index 5, nominal resting length $520\,\mathrm{mm}$) whose femur- and tibia-side contacts engage progressively with flexion. The active-range column lists the knee angles over which each point participates in the path.

Point	Description	Frame	x (mm)	y (mm)	z (mm)	Active θ_k
Flexor, $10\,\mathrm{mm}$ **(two-point route, resting length** $415\,\mathrm{mm}$**)**						
p_1	origin	femur	-50.00	35.00	50.00	all θ_k
p_2	insertion	tibia θ_1	-12.24	-8.87	27.87	all θ_k
Extensor, $10\,\mathrm{mm}$ **(eight-point route, resting length** $520\,\mathrm{mm}$**)**						
p_1	origin	femur	40.00	35.00	0.00	all θ_k
p_2	mounting-base contact	femur	75.90	-274.76	0.00	all θ_k
p_3	femur channel contact	femur	59.82	-374.27	0.00	$\theta_k \le -9.19^\circ$
p_4	femoral condyle contact	femur	39.55	-421.83	0.00	$\theta_k \le -69.8^\circ$
p_5	tibia contact	tibia θ_1	58.71	25.00	0.00	$\theta_k \le -9.19^\circ$
p_6	tibia contact	tibia θ_1	58.71	12.28	0.00	all θ_k
p_7	tibia tendon contact	tibia θ_1	54.00	-6.12	0.00	all θ_k
p_8	patellar ligament ring	tibia θ_1	36.04	-28.44	0.00	all θ_k

Table S5. Routing via points of the $20\,\mathrm{mm}$ route redesigns, in mm. The flexor design routes two BPAs on mirrored paths: the second BPA keeps the z-mirrored distal endpoint while both origins sit off the same side of the knee, and each route carries one tibial wrap point placed at the BPA centerline standoff from the tibial surface while contact is active. The extensor design is the clip-on-cam route; the liftoff column lists the knee angle above which each contact leaves the path as the knee extends, with all contacts engaged at maximum flexion.

Point	Description	Frame	x (mm)	y (mm)	z (mm)	Liftoff θ_k
Flexor, $20\,\mathrm{mm}$ **(two-BPA mirrored route, two BPAs, resting length** $561.49\,\mathrm{mm}$**, tendon** $25.00\,\mathrm{mm}$**)**						
p_1	origin, BPA 1	femur	-125.00	135.00	72.08	end point
p_{end}	insertion, BPA 1	tibia θ_1	-18.72	-32.69	38.96	end point
$p_1^{(2)}$	origin, BPA 2	femur	-125.00	135.00	-5.85	end point
$p_{end}^{(2)}$	insertion, BPA 2	tibia θ_1	-18.72	-32.69	-38.96	end point
Extensor, $20\,\mathrm{mm}$ **(clip-on-cam route, two BPAs, resting length** $827.74\,\mathrm{mm}$**, tendon** $92.49\,\mathrm{mm}$**)**						
p_1	origin (fixed)	femur	40.00	137.00	0.00	end point
p_2	mounting-base contact	femur	83.00	-271.70	0.00	$-21.52°$
p_3	femur contact	femur	66.60	-427.80	0.00	$-54.34°$
p_4	femur contact	femur	54.70	-443.60	0.00	$-89.80°$
p_5	femoral condyle contact	femur	38.80	-454.40	0.00	$-110.81°$
p_6	cam contact (optimized)	tibia θ_1	116.32	15.04	0.00	$-18.89°$
p_7	cam contact (optimized)	tibia θ_1	131.32	-18.97	0.00	none
p_8	cam contact (optimized)	tibia θ_1	127.04	-47.87	0.00	none
p_9	distal ring (insertion)	tibia θ_1	34.42	-197.10	0.00	end point

1.3 Figures

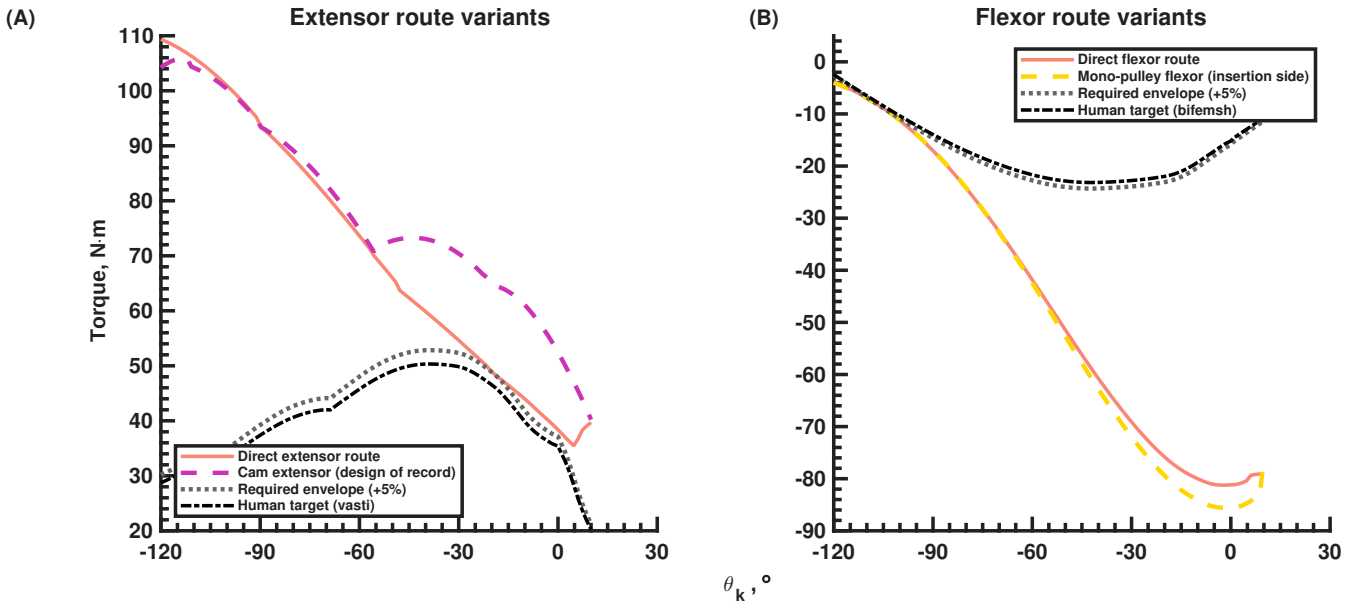


Figure S1: Predicted knee-axis torque of the redesigned route families against the human-scaled requirements. Knee angle is negative in flexion; extension torque is positive and flexion torque negative. **(A)** Extensor routes: the direct route and the cam design of record against the vasti requirement and its required $5\,\%$ envelope. **(B)** Flexor routes: the direct two-BPA route and the mono-pulley variant on the insertion-side measure against the biceps femoris short-head requirement and its envelope.

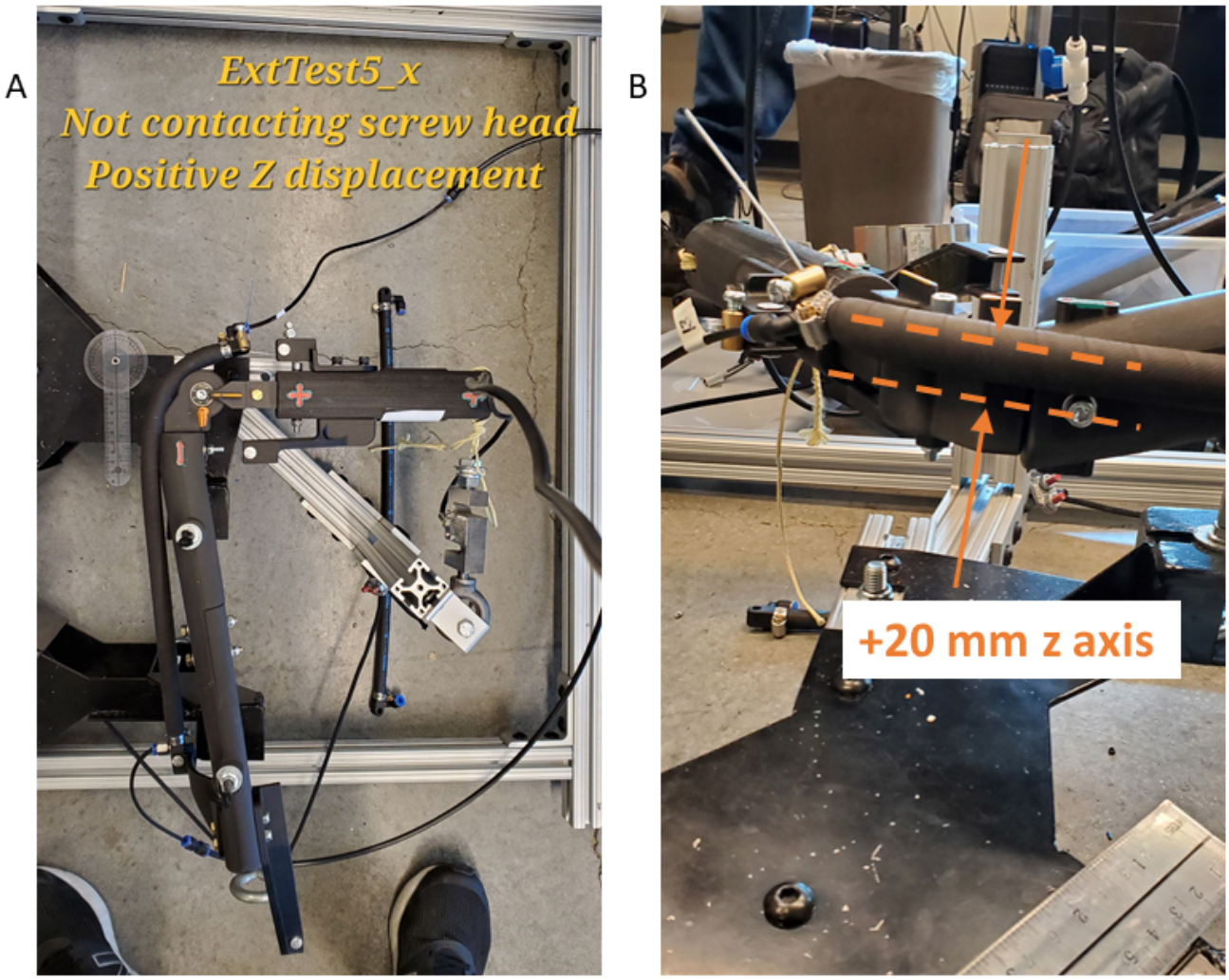


Figure S2: During extensor tests with the pinned knee using a tendon it was noticed that the actuator pathway was modeled to be anterior of the socket head cap screw that joins the femur body to the knee assembly. This slippage affects the muscle path and the moment arm. It was determined this caused a $\pm 20\,\mathrm{mm}\ \hat{z}$ axis displacement. Test assembly is shown with **(A)** top down view and **(B)** side view.